

Synchronized Cardioversion

Synchronized cardioversion can be a life-saving procedure in the setting of a patient with narrow or wide complex tachycardia with evidence of decreased organ perfusion.

Indications for transcutaneous pacing

- Wide complex tachycardia with SBP $\leq$ 90 mmHg and non-responsive to fluid
- Narrow complex tachycardia with SBP $\leq$ 90 mmHg and non-responsive to fluid

Contraindications

- Asystole
- Ventricular fibrillation
- Normal Sinus Rhythm
- Bradycardia

Relative Contraindication

- PEA cardiac arrest – the patient may be in a low flow state. This case can be discussed with Medical Control Physician

Equipment needed

- Cardiac monitor
- Pacer pads
- Consider procedural sedation (Ketamine) or pain management

Synchronized Cardioversion

1. Prepare equipment
 - a. Place pacer pads over
 - i. Anterior/Posterior is preferred
 - ii. Anterior below right clavicle and left lower chest is secondary placement
 - iii. Avoid placement over AICD or transdermal drug patches
2. Activate Synchronize mode – Turn on the monitor and select the synchronized mode
3. Connect cables to the monitor
4. Press the synchronize button and wait for the arrows to indicate the monitor is in sync with the patient's cardiac rhythm
5. Select energy setting
 - a. Initial energy setting 100J (adults) and 0.5 J/kg (child) – max is adult setting
 - b. Repeat energy settings 150J and 200J in adults
 - c. Repeat energy settings 1J/kg and 2 J/kg in child – max is adult setting in stepwise fashion
6. Ensure no one is touching the patient. State "All clear".
7. Push the defibrillate button to deliver the shock
8. Obtain a 12 lead to analyze the rhythm if any changes are noted
9. If no changes are noted, repeat steps 4-8 at increasing energy settings